manure allocation

Purpose

`manure_allo.mnu` defines manure-allocation objects. Each object names one or more source locations (manure storage on a farm, a feedlot, a lagoon) and one or more demand locations (HRUs, municipal treatment, inter-basin diversion). At runtime, manure produced at the sources is distributed to the demand objects under a named rule. Each allocation object holds its own source list, demand list, monthly production, storage, and per-source withdrawal history.

The allocation file ties three databases together:

- `fertilizer.frt`. Each source's `manure_typ` is a key into `fertdb` (cross-walked at read time into `src%fertdb`).
- `lum.dtl`. For HRU demands, the rule name is a key into `dtbl_lum` (cross-walked into `trn%dtbl_num`).
- `chem_app.ops`. The first action of the matched lum decision table names a chemical-application profile in `chemapp_db`, cross-walked into `manure_amt%app_method`.

Source

- Reader: `src/manure_allocation_read.f90`
- Types: `src/manure_allocation_module.f90`
  - `type manure_allocation` (variable `mallo`)
  - `type manure_source_objects` (`mallo(i)%src(j)`)
  - `type manure_demand_objects` (`mallo(i)%trn(j)`)

Format

- Line 1: title (skipped).
- Line 2: number of allocation objects, `imax`. Stored in `db_mx%mallo_db`.
- Per allocation object, in order:
  1. Header line (skipped).
  2. Object header record: `name`, `rule_typ`, `src_obs`, `trn_obs`.
  3. Header line (skipped).
  4. `src_obs` source records.
  5. Header line (skipped).
  6. `trn_obs` demand records.

Object header

#	Field	Type	Description
1	name	char(25)	Allocation-object name.
2	rule_typ	char(25)	Allocation rule. Selects how unmet demand is split across sources.
3	src_obs	int	Number of source objects in this allocation.
4	trn_obs	int	Number of demand objects in this allocation.

Source record (one per source)

#	Field	Type	Units	Description
1	num	int	none	Sequential source id. Used as the array index.
2	mois_typ	char(3)	none	Moisture class: `wet` or `dry`.
3	manure_typ	char(25)	none	Manure-product name. Cross-walked against `fertdb(:)%fertnm` and stored in `fertdb`.
4	lat	real	degrees	Latitude of the source.
5	long	real	degrees	Longitude of the source.
6	stor_init	real	t	Initial manure in storage.
7	stor_max	real	t	Storage capacity.
8..19	prod_mon(1..12)	real	t/month	Average monthly manure production. Twelve values, January through December.

Demand record (one per demand)

#	Field	Type	Description
1	num	int	Sequential demand id. Used as the array index.
2	ob_typ	char(10)	Receiver type: `hru` (apply to an HRU), `muni` (municipal treatment), or `divert` (inter-basin diversion).
3	ob_num	int	Number of the receiving object of `ob_typ`.
4	dtbl	char(25)	Decision-table name. For `hru`, matched against `dtbl_lum`.
5	right	char(2)	Allocation right. `sr` for senior, `jr` for junior.

For an HRU demand, the reader sets `hru(ihru)%man_trn_dtbl` to the resolved decision-table index, and looks up the first action's `option` against `chemapp_db(:)%name` to set the application method on `mallo(imro)%trn(itrn)%manure_amt%app_method`.

Example

A two-line skeleton with one source and one HRU demand:

```
manure_allo.mnu: example
1
name            rule_typ    src_obs  trn_obs
farm_a          fixed             1        1
num  mois_typ  manure_typ  lat       long      stor_init  stor_max   prod_mon(1..12)
  1  wet       beef_salt   41.99   -93.62          0.0       500.0    5.0  5.0  6.0  6.0  7.0  7.0  7.0  7
num  ob_typ    ob_num  dtbl                 right
  1  hru            1  apply_manure_dtbl       sr
```

The decision-table named `apply_manure_dtbl` must exist in `lum.dtl` and its first action must name an entry in `chem_app.ops` for the cross-walks to resolve.

Related files

- `fertilizer.frt`. Defines the manure-product names that source records reference, and the manure organic-matter database (`manure_db.frt`, `manure_om.frt`).
- `lum.dtl`. Decision tables that fire manure applications at each demand HRU.
- `chem_app.ops`. Chemical-application profiles (rate, method) selected by the decision-table action.